

Suspicious Windows Endpoint Investigation

Sysmon and Splunk Cloud Case Study

Prepared by	Environment	Report date
Benjamin Alvarez	WIN11-LAB01	27 September 2026

This case study documents the construction and investigation of a Windows 11 laboratory endpoint using Sysmon telemetry and Splunk Cloud. The investigation confirmed two controlled process chains: PowerShell launching Notepad and PowerShell launching cmd.exe to create a marker file. The observed activity was generated deliberately for validation, and the available evidence does not establish an unauthorized compromise.

[Public portfolio edition](#) | [Controlled laboratory activity](#)

Executive Summary

The objective was to validate endpoint telemetry, ingest it into Splunk Cloud, and reconstruct process lineage from the resulting events. Sysmon recorded process creation with command lines, hashes, integrity level, user context, and parent process identifiers. Splunk searches then isolated the relevant events and confirmed the parent child relationships.

Key Findings

- WIN11-LAB01 ran Windows 11 Pro on an ARM64 VMware virtual machine with UEFI Secure Boot enabled.
- Sysmon v15.22 ARM64 and the Sysmon driver were installed and running.
- The initial Splunk dataset contained 89 events, including 53 Sysmon Event ID 1 process creation records.
- PowerShell PID 8536 launched Notepad PID 13424, which then launched the packaged Notepad process PID 8768.
- PowerShell PID 8536 launched cmd.exe PID 12144 with a command that wrote CaseStudy to C:\Users\Public\marker.txt.
- Two PowerShell temporary policy test file events were unrelated to the controlled marker and were excluded from the primary finding.

Assessment

The process chains are fully consistent with the controlled actions performed during the lab. PowerShell and cmd.exe are legitimate administrative tools that also appear in adversary activity, so the commands, parent processes, user context, timestamps, and resulting artifacts were evaluated together. Within the evidence selected for this case, no corroborating behavior demonstrated persistence, credential access, data exfiltration, or command and control.

Evidence Confidence

Finding	Confidence	Basis
Telemetry collection operated	High	Running Sysmon service and driver plus recent event records
Notepad lineage reconstructed	High	Parent and child identifiers in Sysmon Event ID 1 records
Marker command reconstructed	High	Full cmd.exe command line and PowerShell parent identifiers
No confirmed compromise	Moderate	Controlled context and no corroborating malicious behavior in the supplied snapshots

Laboratory Environment

The endpoint was built as a controlled training system named WIN11-LAB01. A clean baseline snapshot was captured before telemetry configuration so the system could be restored after testing.

Component	Configuration
Guest operating system	Microsoft Windows 11 Pro build 26200
Architecture	ARM64 based PC
Virtualization	VMware Fusion virtual machine
Firmware	UEFI with Secure Boot enabled
Network	NAT through the host connection
Endpoint telemetry	Microsoft Sysmon v15.22 ARM64
Analysis platform	Splunk Cloud Search and Reporting

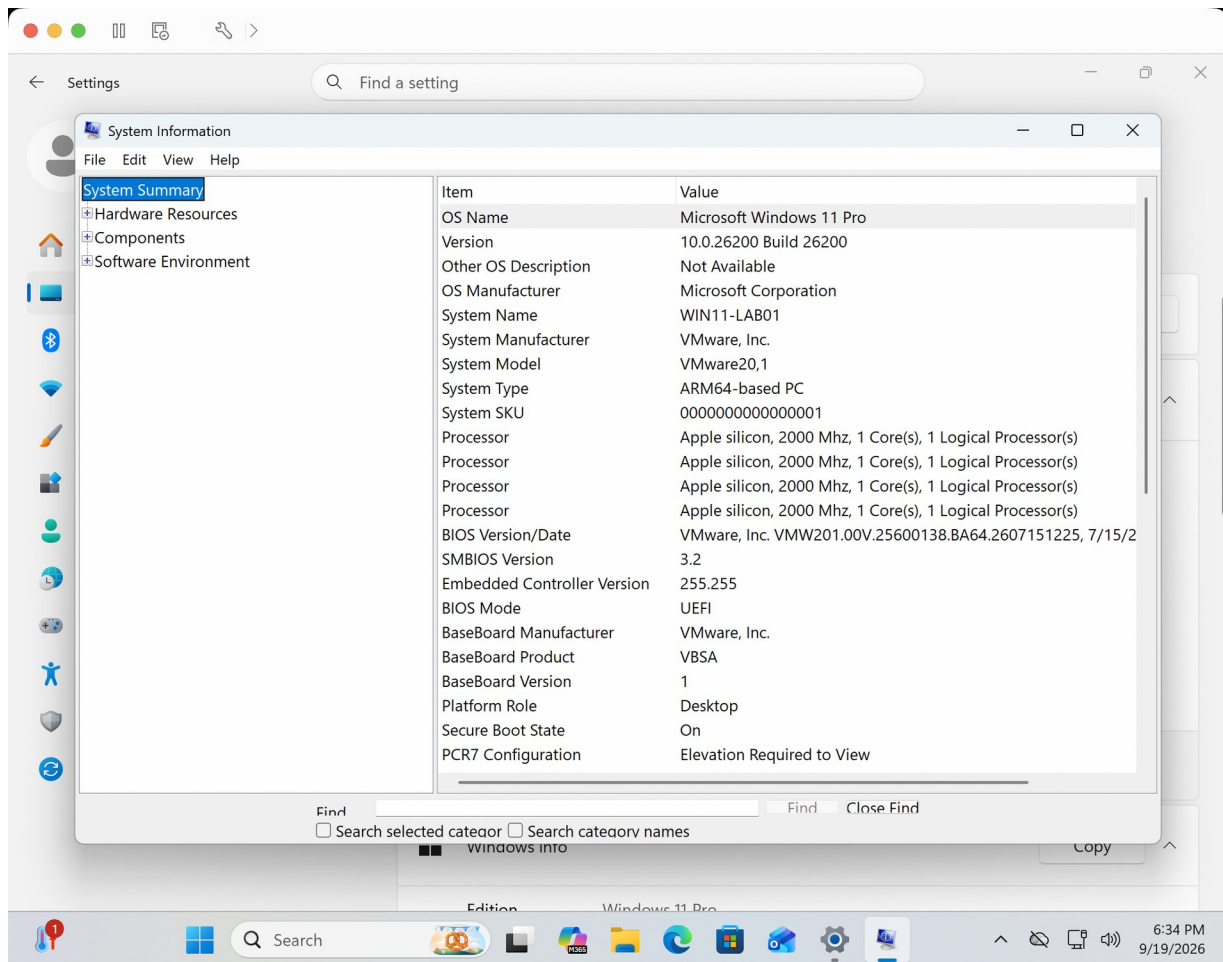


Figure 1 WIN11-LAB01 system summary Evidence EV-VM-009

Telemetry Configuration and Validation

Sysmon was installed from the Microsoft Sysinternals package using the ARM64 executable. The configuration loaded successfully, and both the Sysmon64a service and SysmonDrv driver reported a running state. Event validation confirmed that the Microsoft Windows Sysmon Operational log contained recent records, including Event ID 1 process creation events.

The investigation used manual CSV exports that were uploaded to Splunk Cloud. The first export contained 89 events and supported the Notepad lineage exercise. A later export captured the cmd.exe marker test. The embedded Sysmon Message content was treated as the authoritative event narrative because generic CSV serialization also included .NET event object metadata that did not always describe the embedded event cleanly.

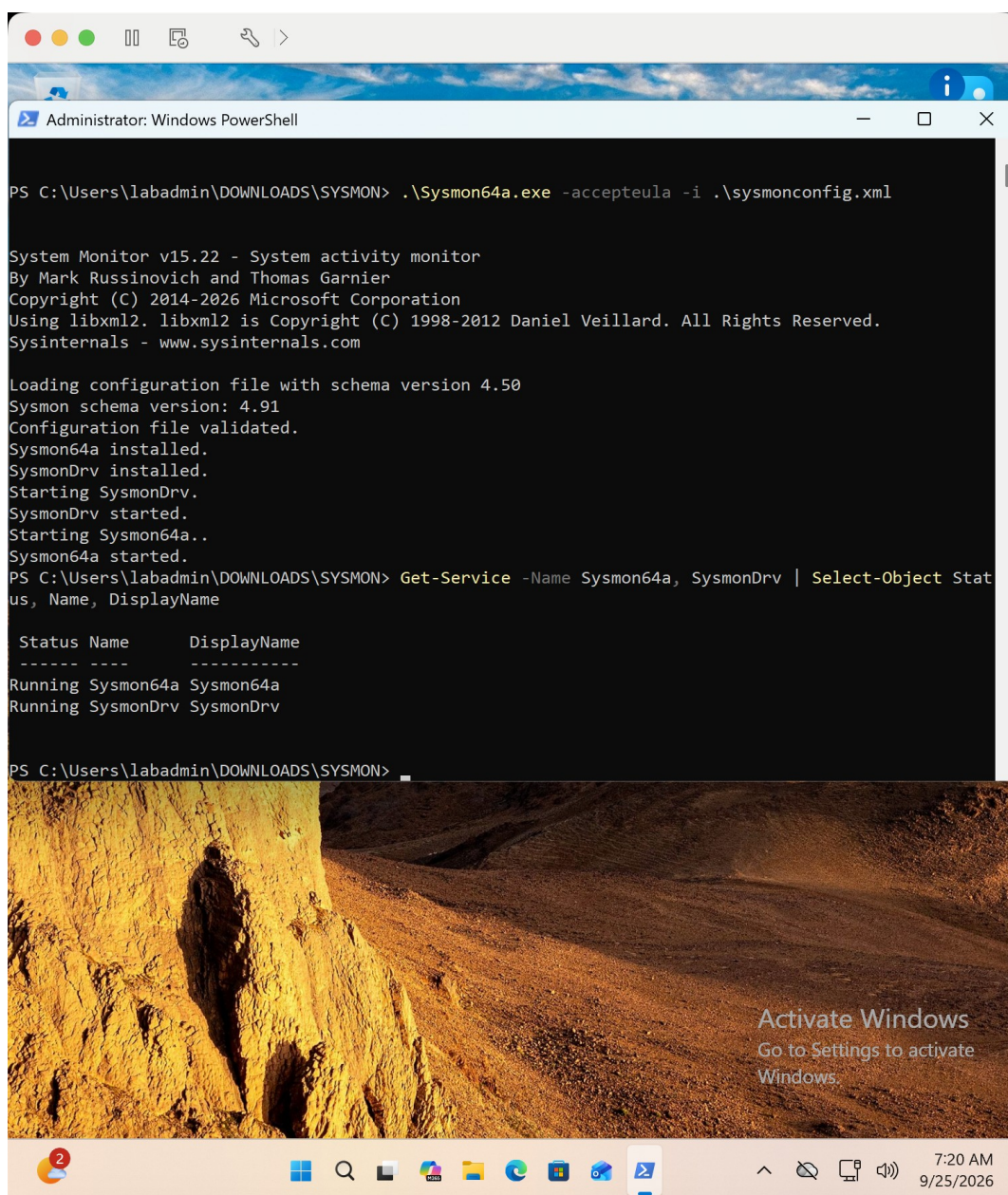


Figure 2 Sysmon service and driver running Evidence EV-SYS-004

Investigation Method

The analysis started with a broad host and source search, then narrowed to process creation records and specific command line terms. Each candidate event was expanded to inspect the process ID, image path, command line, user, integrity level, hashes, parent process ID, and parent image.

Step	Action	Purpose
1	Validate the uploaded source and host	Confirm that WIN11-LAB01 data was searchable
2	Filter for Sysmon Event ID 1	Create the process creation working set
3	Search for notepad.exe	Isolate the controlled Notepad activity
4	Inspect parent and child identifiers	Reconstruct PowerShell to Notepad lineage
5	Search for marker.txt	Locate the controlled file creation command
6	Validate the cmd.exe parent	Confirm PowerShell PID 8536 as the launcher
7	Review nearby PowerShell file events	Exclude routine temporary policy test files

Process Lineage

Step	Process or artifact	Relationship	Evidence
1	powershell.exe PID 8536	Parent shell	EV-INV-003 and EV-INV-005
2	notepad.exe PID 13424	Child of PowerShell	EV-INV-003
3	Notepad.exe PID 8768	Child of Notepad PID 13424	EV-INV-004
1	powershell.exe PID 8536	Parent shell	EV-INV-005
2	cmd.exe PID 12144	Child of PowerShell	EV-INV-005
3	C:\Users\Public\marker.txt	Artifact created by command	EV-INV-005 and local verification

Investigation Timeline

Times below use UTC when the Sysmon UtcTime value was available. The Windows interface displayed Pacific daylight time, seven hours behind UTC on the investigation dates.

Time	Event	Finding	Evidence
2026 09 25 16 47 22.996 UTC	Process Create	PowerShell PID 8536 launched Notepad PID 13424	EV-INV-003
2026 09 25 16 47 23.089 UTC	Process Create	Notepad PID 13424 launched packaged Notepad PID 8768	EV-INV-004
2026 09 27 14 46 08.129 UTC	Process Create	PowerShell PID 8536 launched cmd.exe PID 12144	EV-INV-005
2026 09 27 14 46 08.129 UTC	Command line	cmd.exe wrote CaseStudy to C:\Users\Public\marker.txt	EV-INV-005

Decisive Evidence

Sysmon Event ID 1 recorded the complete marker command, the executing user, high integrity, image hash values, and the PowerShell parent. The local verification command then read the marker file and returned the expected text CaseStudy. Together, these observations connect the parent process, child process, command, and resulting artifact.

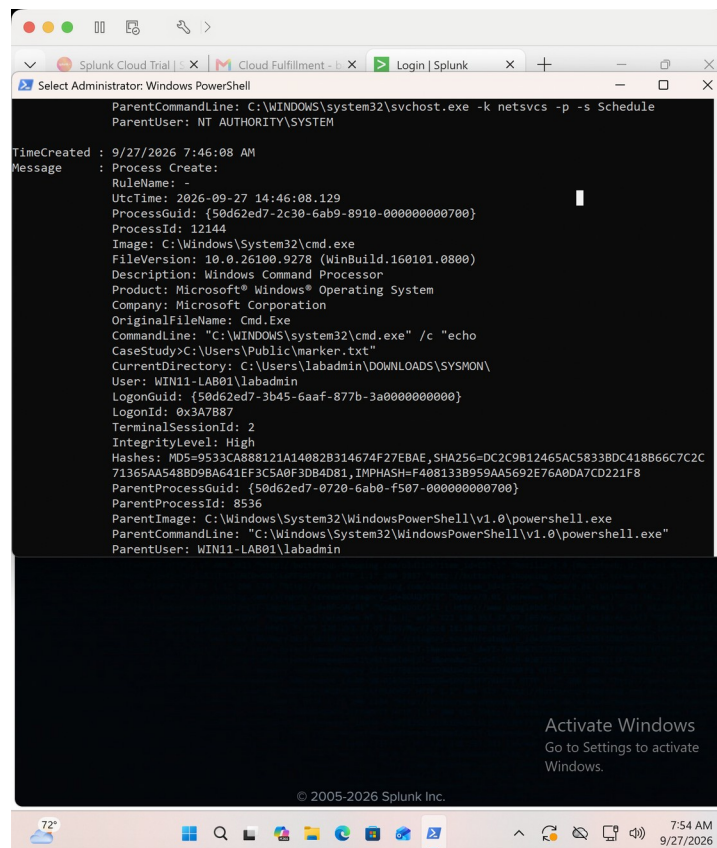


Figure 3 cmd.exe marker command and PowerShell parent Evidence EV-INV-005

Technical Assessment

Supported Conclusion

The investigated process activity was produced by the analyst during controlled telemetry validation. The command lines, timestamps, account, parent child relationships, and marker contents agree with the documented test steps. The evidence therefore supports a benign lab classification for the selected events.

Alternative Explanations

PowerShell and cmd.exe can support malicious execution, but executable names alone do not establish intent. The exact marker command and resulting content match the controlled test. Separate PowerShell file events referenced __PSScriptPolicyTest temporary files in the lab user's Temp directory. Those files are routine PowerShell policy checks and did not match the marker path, so they were excluded from the main process chain.

MITRE ATT&CK Context

Technique	Observed behavior	Interpretation
T1059.001 PowerShell	PowerShell served as the parent shell for controlled process launches	Technique relevant for detection engineering but benign in this test
T1059.003 Windows Command Shell	cmd.exe executed an echo command that created marker.txt	Command shell use was deliberate and locally verified

Limitations

- The dataset consisted of manual CSV snapshots rather than continuous endpoint forwarding.
- The process creation record for the long running PowerShell PID 8536 was not present in the imported snapshot, so the supported lineage begins with that observed parent.
- No memory image, full disk image, packet capture, or enterprise identity telemetry was collected.
- The conclusion applies to the selected events and should not be generalized beyond the available evidence.

Recommendations

- Forward Windows event data continuously with the Splunk Universal Forwarder or another supported collector.
- Use structured Windows event fields or a supported Splunk technology add on instead of generic CSV serialization.
- Normalize timestamps to UTC and preserve the endpoint timezone in case notes.
- Retain ProcessGuid and ParentProcessGuid values to correlate activity when Windows process identifiers are reused.

Evidence Index

The following evidence items support the conclusions in this report. Original screenshots and SHA256 values are maintained in the case evidence register.

Evidence	Description	Use in report
EV-VM-009	WIN11-LAB01 Windows system summary	Laboratory environment
EV-SYS-004	Sysmon64a service and SysmonDrv driver running	Telemetry configuration
EV-TEL-001	Recent Sysmon events displayed in PowerShell	Telemetry validation
EV-TEL-006	Initial Splunk search returned 89 events	Ingestion validation
EV-INV-001	Fifty three process creation events isolated	Investigation baseline
EV-INV-003	Notepad PID 13424 with PowerShell parent PID 8536	Notepad lineage
EV-INV-004	Packaged Notepad PID 8768 with parent PID 13424	Notepad lineage
EV-INV-005	cmd.exe PID 12144 with marker command and PowerShell parent	Marker lineage

Final Disposition

Disposition Benign controlled laboratory activity. The case demonstrates successful endpoint telemetry collection, Splunk ingestion, targeted searching, process lineage reconstruction, evidence handling, and defensible scoping of conclusions.